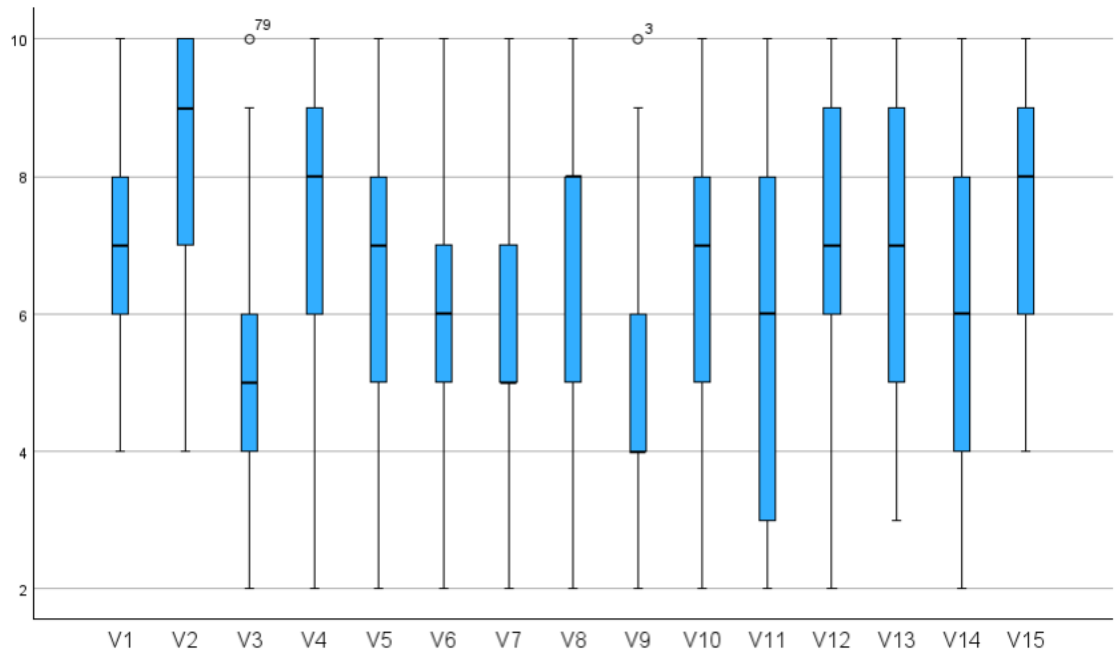
Favorite Location – Paris Jalloh

This study investigates the latent factors that shape individuals' preferences for their favorite locations in Paris. To accomplish this, a factor analysis is used to reduce the dimensionality of the data and uncover key variation in preferences.

Model 1 – All



The difference in variation between variables shows possible dimensionality where factor analysis could be a useful tool to remove.

Correlation Matrix

		V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15
Correlation	V1	1.000	.040	-.055	-.023	-.282	.105	.048	-.255	.228	.128	.080	-.039	-.209	-.064	-.166
	V2	.040	1.000	-.361	.675	-.068	.039	.049	.362	-.075	-.100	.238	.651	.002	-.031	-.141
	V3	-.055	-.361	1.000	-.133	.173	.186	-.110	-.400	-.107	.057	-.080	-.113	-.048	.125	.003
	V4	-.023	.675	-.133	1.000	.013	.116	.008	.394	.065	.023	.082	.750	-.091	-.078	-.155
	V5	-.282	-.068	.173	.013	1.000	.331	.207	-.035	-.191	-.117	-.040	.068	.062	.165	-.001

V6	.105	.039	.186	.116	.331	1.000	.183	.017	.253	-.215	.194	.102	.078	-.052	.126
V7	.048	.049	-.110	.008	.207	.183	1.000	.079	.052	.033	-.022	.130	.026	.065	-.094
V8	-.255	.362	-.400	.394	-.035	.017	.079	1.000	.035	-.095	-.027	.364	.109	-.107	.024
V9	.228	-.075	-.107	.065	-.191	.253	.052	.035	1.000	.010	-.039	.082	-.111	.056	-.136
V10	.128	-.100	.057	.023	-.117	-.215	.033	-.095	.010	1.000	-.183	-.113	-.120	.100	-.244
V11	.080	.238	-.080	.082	-.040	.194	-.022	-.027	-.039	-.183	1.000	.024	.125	-.216	.184
V12	-.039	.651	-.113	.750	.068	.102	.130	.364	.082	-.113	.024	1.000	-.136	.029	-.186
V13	-.209	.002	-.048	-.091	.062	.078	.026	.109	-.111	-.120	.125	-.136	1.000	-.352	.415
V14	-.064	-.031	.125	-.078	.165	-.052	.065	-.107	.056	.100	-.216	.029	-.352	1.000	-.315
V15	-.166	-.141	.003	-.155	-.001	.126	-.094	.024	-.136	-.244	.184	-.186	.415	-.315	1.000

Overall, the correlation matrix shows that the variables are well correlated therefore we can continue with the extraction of latent factors.

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.583
Bartlett's Test of Sphericity	Approx. Chi-Square	319.724
	df	105
	Sig.	<.001

Communalities

	Initial	Extraction
V1	1.000	.731
V2	1.000	.804
V3	1.000	.662

V4	1.000	.789
V5	1.000	.727
V6	1.000	.777
V7	1.000	.812
V8	1.000	.706
V9	1.000	.816
V10	1.000	.461
V11	1.000	.576
V12	1.000	.789
V13	1.000	.571
V14	1.000	.538
V15	1.000	.611

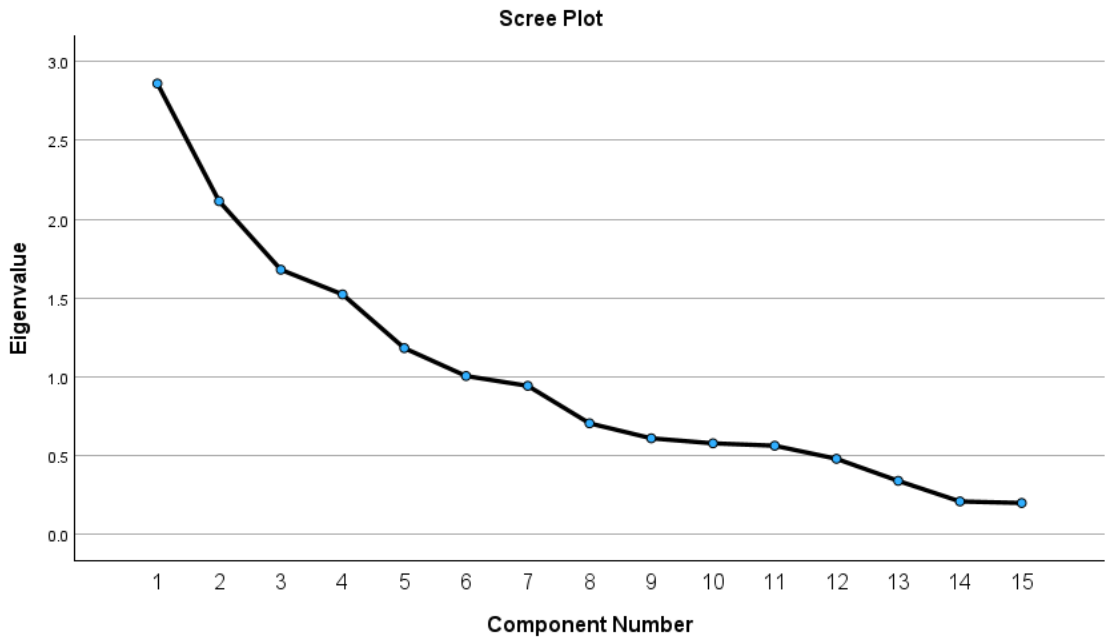
Extraction Method: Principal Component Analysis.

Total Variance Explained

	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
Component	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.862	19.077	19.077	2.862	19.077	19.077	2.658	17.723	17.723
2	2.115	14.100	33.177	2.115	14.100	33.177	2.036	13.571	31.294
3	1.680	11.202	44.379	1.680	11.202	44.379	1.553	10.356	41.649
4	1.523	10.156	54.535	1.523	10.156	54.535	1.532	10.212	51.861
5	1.183	7.884	62.419	1.183	7.884	62.419	1.303	8.687	60.548
6	1.006	6.704	69.123	1.006	6.704	69.123	1.286	8.575	69.123
7	.943	6.289	75.412						
8	.705	4.703	80.115						
9	.610	4.069	84.184						
10	.579	3.857	88.041						
11	.563	3.755	91.796						
12	.480	3.202	94.998						
13	.340	2.270	97.268						

14	.210	1.399	98.667						
15	.200	1.333	100.000						

Extraction Method: Principal Component Analysis.



Rotated Component Matrix[a]

	Component					
	1	2	3	4	5	6
V1	-.018	-.073	.000	.839	.145	.028
V2	.863	.089	-.177	.090	-.105	.033
V3	-.190	-.132	.768	-.074	.025	-.108
V4	.881	-.064	-.071	-.018	.053	-.032
V5	.057	-.031	.458	-.462	.029	.547
V6	.141	.217	.378	.036	.652	.376
V7	.010	-.063	-.175	.081	.029	.877
V8	.426	.080	-.582	-.401	.128	.048
V9	-.063	-.221	-.272	.251	.789	-.053
V10	-.135	-.383	-.138	.298	-.418	.114
V11	.266	.571	.221	.360	-.008	.029
V12	.861	-.145	-.047	-.089	.119	.054
V13	-.140	.667	-.151	-.242	-.060	.148
V14	-.032	-.678	.193	-.177	.032	.086

V15	-.207	.704	.021	-.231	.088	-.105

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.
a. Rotation converged in 12 iterations.

The KMO value (.583) indicated mild sampling adequacy, while Bartlett's test was significant with a p value below 0.01. Six factors with eigenvalues greater than 1 were found, which explain 69.1% of the variance. The rotated component matrix shows several variables (V5, V6, V8, V10) with significant cross-loadings or weak loadings, which were removed.

Model 2 – Without V5, V6, V8, V10

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.604
Bartlett's Test of Sphericity	Approx. Chi-Square	212.970
	df	55
	Sig.	<.001

Communalities

	Initial	Extraction
V1	1.000	.681
V2	1.000	.797
V3	1.000	.471
V4	1.000	.803
V7	1.000	.700
V9	1.000	.425
V11	1.000	.444
V12	1.000	.788
V13	1.000	.628
V14	1.000	.559
V15	1.000	.576

Extraction Method: Principal Component Analysis.

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.580	23.452	23.452	2.580	23.452	23.452	2.508	22.804	22.804
2	1.924	17.490	40.942	1.924	17.490	40.942	1.841	16.735	39.539
3	1.305	11.861	52.803	1.305	11.861	52.803	1.401	12.737	52.275
4	1.065	9.679	62.482	1.065	9.679	62.482	1.123	10.207	62.482
5	.913	8.297	70.779						
6	.863	7.849	78.628						
7	.763	6.937	85.565						
8	.576	5.237	90.802						
9	.557	5.068	95.870						
10	.233	2.117	97.987						
11	.221	2.013	100.000						

Extraction Method: Principal Component Analysis.

Rotated Component Matrix[a]

	Component			
	1	2	3	4
V1	-.034	.046	.822	-.054
V2	.877	.124	.027	.106

V3	-.263	-.255	-.234	-.531
V4	.896	-.013	-.008	-.026
V7	.003	-.151	.002	.823
V9	-.023	-.099	.608	.214
V11	.199	.573	.216	-.169
V12	.871	-.141	-.039	.093
V13	-.129	.630	-.388	.254
V14	-.035	-.744	-.065	-.002
V15	-.223	.654	-.314	-.019

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.
a. Rotation converged in 6 iterations.

After removing variables with cross-loadings (V5, V6, V8, V10), the factor analysis was rerun. The KMO value improved to .604, showing acceptable sampling adequacy, and Bartlett's test remained significant. Four factors with eigenvalues greater than 1 were retained, explaining 62.48% of the total variance. Variables (V3, V9, V11) with relatively low communalities were removed, and a final model was estimated to achieve a cleaner and more interpretable factor structure. (V3, V9, V11) with relatively low communalities were removed.

Model 3 – Without V3, V9, V11

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.667
Bartlett's Test of Sphericity	Approx. Chi-Square	172.316
	df	28
	Sig.	<.001

Communalities

	Initial	Extraction
V1	1.000	.893
V2	1.000	.748
V4	1.000	.836

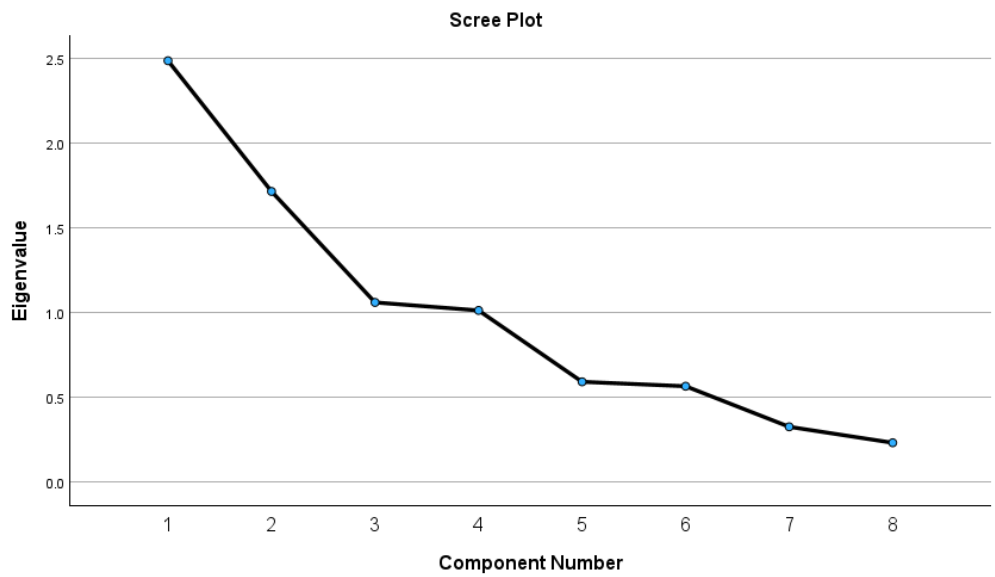
V7	1.000	.975
V12	1.000	.817
V13	1.000	.683
V14	1.000	.727
V15	1.000	.602

Extraction Method: Principal Component Analysis.

Total Variance Explained

	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
Component	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.489	31.107	31.107	2.489	31.107	31.107	2.412	30.154	30.154
2	1.717	21.459	52.566	1.717	21.459	52.566	1.745	21.814	51.968
3	1.061	13.268	65.833	1.061	13.268	65.833	1.095	13.683	65.650
4	1.014	12.678	78.511	1.014	12.678	78.511	1.029	12.861	78.511
5	.593	7.408	85.919						
6	.567	7.082	93.002						
7	.327	4.088	97.090						
8	.233	2.910	100.000						

Extraction Method: Principal Component Analysis.



Rotated Component Matrix[a]

	Component			
	1	2	3	4
V1	-.032	-.125	.935	.042
V2	.863	.008	.044	.025
V4	.912	-.030	.003	-.055
V7	.044	-.028	.034	.985
V12	.888	-.124	-.057	.101
V13	-.049	.780	-.227	.144
V14	-.106	-.760	-.358	.102
V15	-.172	.726	-.186	-.105

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.
a. Rotation converged in 5 iterations.

In the final model, after removing variables with low communalities (V3, V9, V11), the factor analysis produced a better model. The KMO measure increased to .667, indicating acceptable sampling adequacy, while Bartlett's test remained significant. Four factors with eigenvalues greater than 1 were retained, explaining 78.51% of the total variance, representing a substantial improvement from previous models.

The rotated component matrix showed that all variables loaded strongly onto a single factor, with no significant cross-loadings. Factor 1 consisted of V2, V4, and V12; Factor 2 included V13, V14, and V15; Factor 3 was defined by V1; and Factor 4 was defined by V7.

Conclusion:

The final model demonstrated improved sampling adequacy, higher explained variance, and a clear factor structure with strong loadings and no cross-loadings. This suggests that the retained variables effectively capture the underlying dimensions of the data, resulting in a robust and parsimonious factor solution.